

DATA SHEET

For a complete data sheet, please also download:

- The IC04 LOCMOS HE4000B Logic Family Specifications HEF, HEC
- The IC04 LOCMOS HE4000B Logic Package Outlines/Information HEF, HEC

HEF4511B **MSI** BCD to 7-segment latch/decoder/driver

Product specification
File under Integrated Circuits, IC04

January 1995

BCD to 7-segment latch/decoder/driver

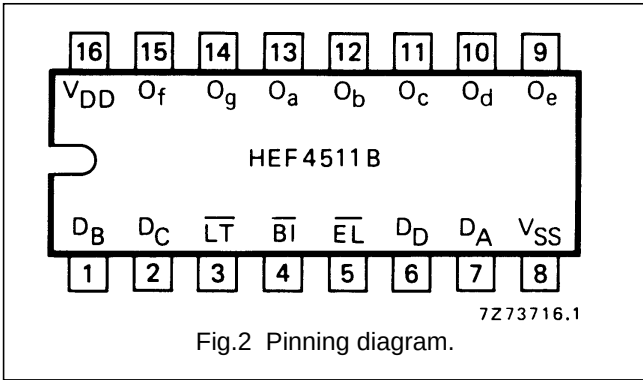
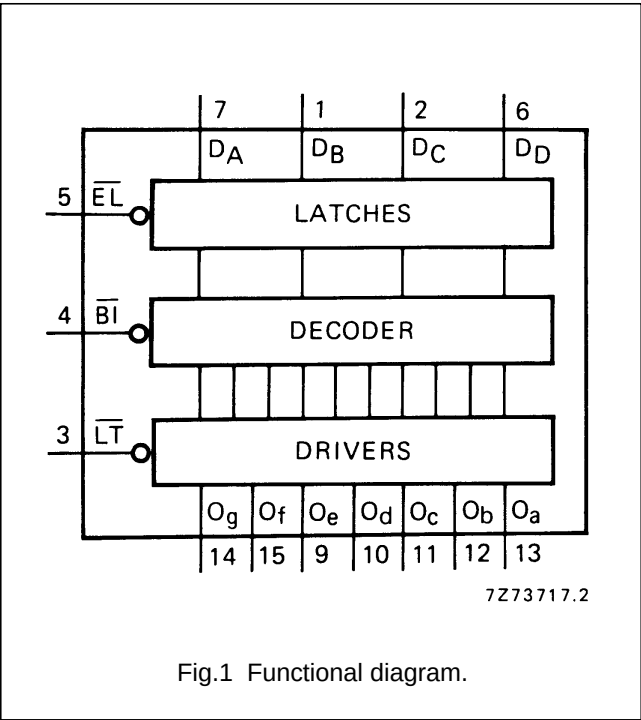
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DESCRIPTION

The HEF4511B is a BCD to 7-segment latch/decoder/driver with four address inputs (D_A to D_D), an active LOW latch enable input ($\overline{EL}$), an active LOW ripple blanking input ($\overline{BI}$), an active LOW lamp test input ($\overline{LT}$), and seven active HIGH n-p-n bipolar transistor segment outputs (O_a to O_g).

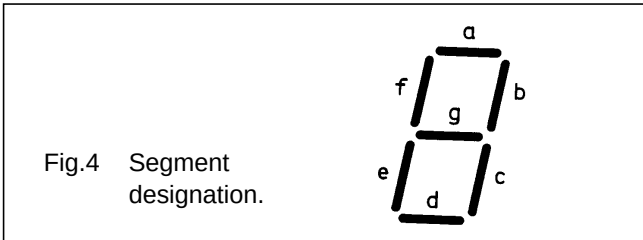
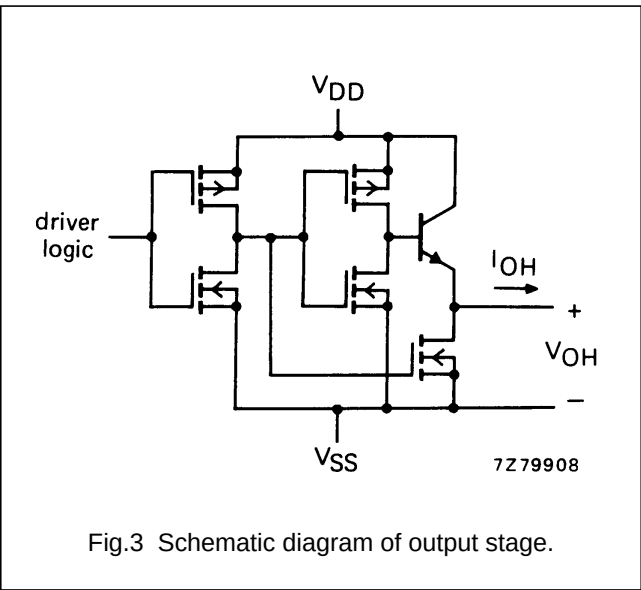
When $\overline{EL}$ is LOW, the state of the segment outputs (O_a to O_g) is determined by the data on D_A to D_D . When $\overline{EL}$ goes HIGH, the last data present on D_A to D_D are stored in the latches and the segment outputs remain stable. When $\overline{LT}$ is LOW, all the segment outputs are HIGH independent of all other input conditions. With $\overline{LT}$ HIGH, a LOW on $\overline{BI}$ forces all segment outputs LOW. The inputs $\overline{LT}$ and $\overline{BI}$ do not affect the latch circuit.



- HEF4511BP(N): 16-lead DIL; plastic (SOT38-1)
- HEF4511BD(F): 16-lead DIL; ceramic (cerdip) (SOT74)
- HEF4511BT(D): 16-lead SO; plastic (SOT109-1)
- (): Package Designator North America

PINNING

- D_A to D_D address (data) inputs
- $\overline{EL}$ latch enable input (active LOW)
- $\overline{BI}$ ripple blanking input (active LOW)
- $\overline{LT}$ lamp test input (active LOW)
- O_a to O_g segment outputs



FAMILY DATA, I_{DD} LIMITS category MSI

See Family Specifications

BCD to 7-segment latch/decoder/driver

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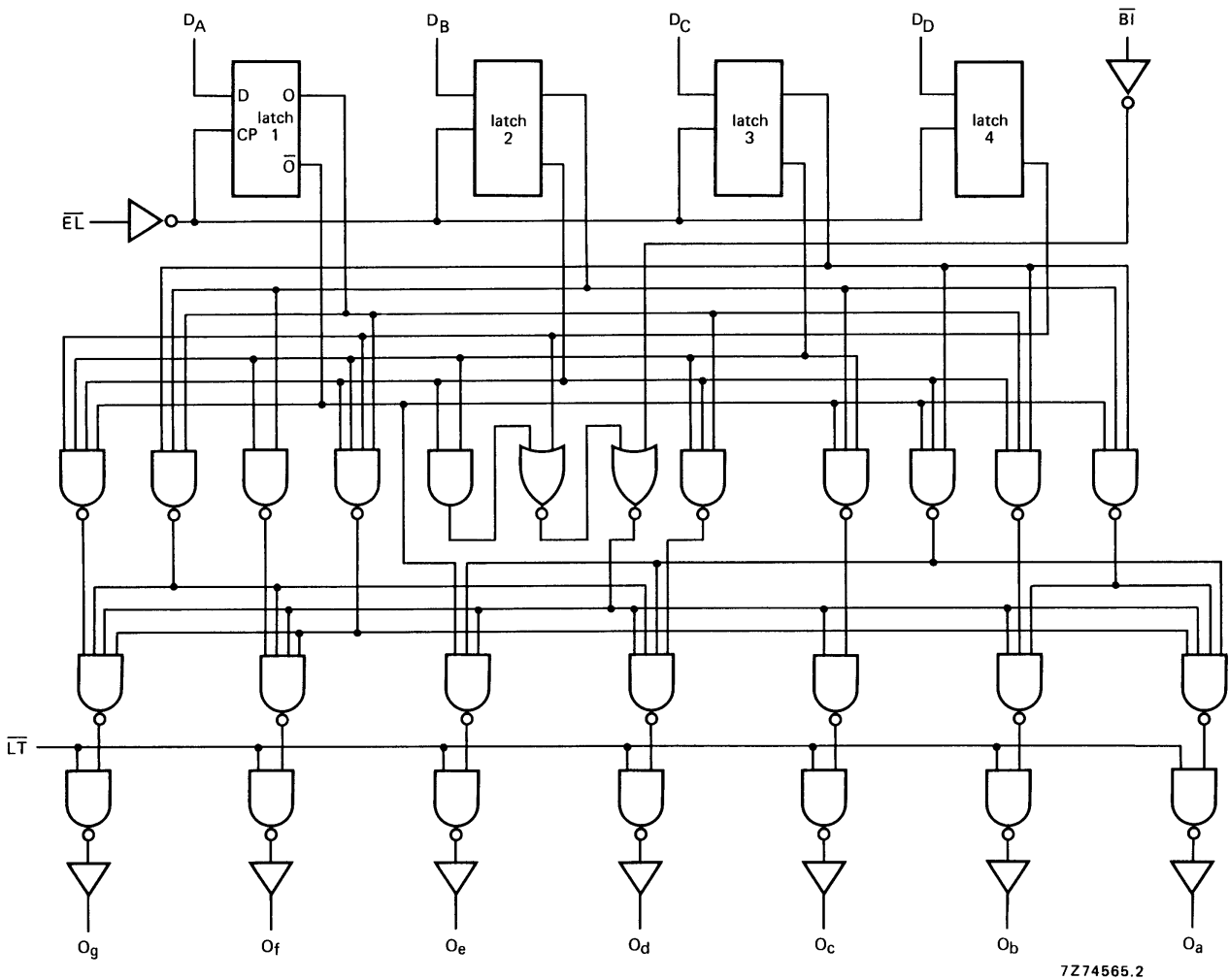
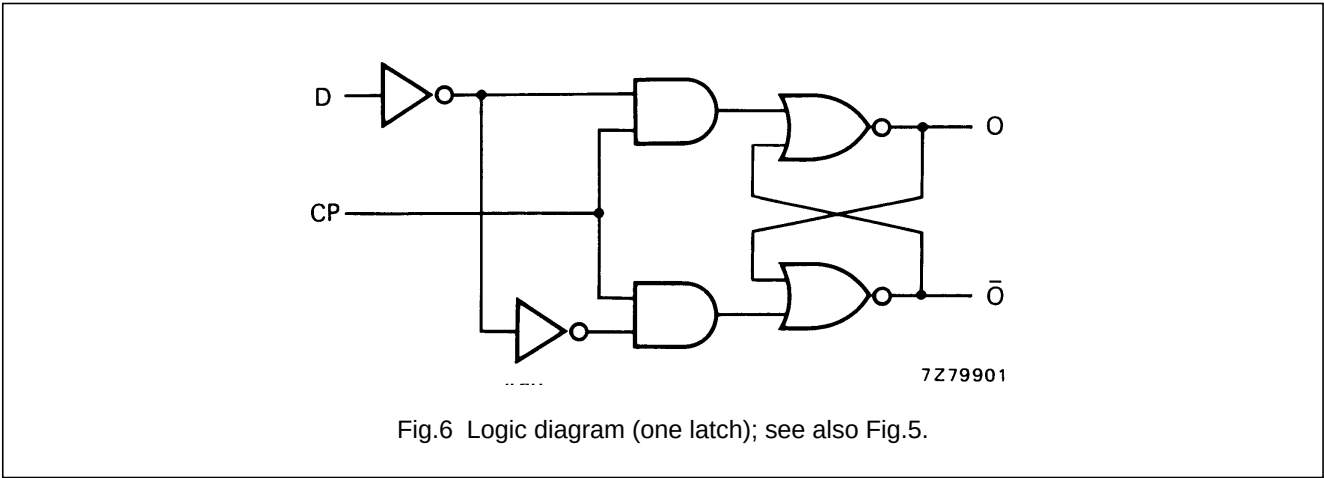


Fig.5 Logic diagram; for one latch see Fig.6.

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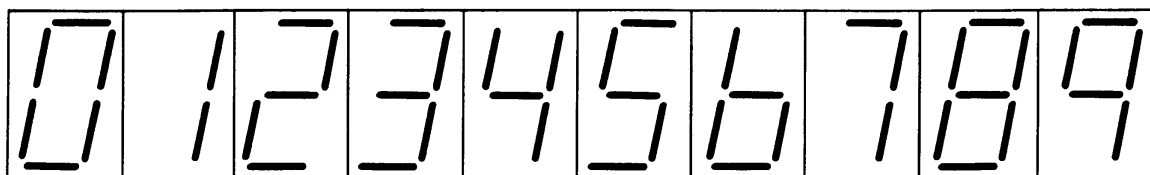
FUNCTION TABLE

INPUTS							OUTPUTS							DISPLAY
$\overline{\text{EL}}$	$\overline{\text{BI}}$	$\overline{\text{LT}}$	D_D	D_C	D_B	D_A	O_a	O_b	O_c	O_d	O_e	O_f	O_g	
X	X	L	X	X	X	X	H	H	H	H	H	H	H	8
X	L	H	X	X	X	X	L	L	L	L	L	L	L	blank
L	H	H	L	L	L	L	H	H	H	H	H	H	L	0
L	H	H	L	L	L	H	L	H	H	L	L	L	L	1
L	H	H	L	L	H	L	H	H	L	H	H	L	H	2
L	H	H	L	L	H	H	H	H	H	H	L	L	H	3
L	H	H	L	H	L	L	L	H	H	L	L	H	H	4
L	H	H	L	H	L	H	H	L	H	H	L	H	H	5
L	H	H	L	H	H	L	L	L	H	H	H	H	H	6
L	H	H	L	H	H	H	H	H	H	L	L	L	L	7
L	H	H	H	L	L	L	H	H	H	H	H	H	H	8
L	H	H	H	L	L	H	H	H	H	L	L	H	H	9
L	H	H	H	L	H	L	L	L	L	L	L	L	L	blank
L	H	H	H	L	H	H	L	L	L	L	L	L	L	blank
L	H	H	H	H	L	L	L	L	L	L	L	L	L	blank
L	H	H	H	H	H	L	L	L	L	L	L	L	L	blank
L	H	H	H	H	H	H	L	L	L	L	L	L	L	blank
H	H	H	X	X	X	X	*							*

Note

1. H = HIGH state (the more positive voltage)
L = LOW state (the less positive voltage)
X = state is immaterial
* Depends upon the BCD code applied during the LOW to HIGH transition of $\overline{\text{EL}}$.

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Fig.7 Display.

RATINGS

Limiting values in accordance with the Absolute Maximum System (IEC 134).

Output (source) current HIGH $-I_{OH}$ max. 25 mA

For other RATINGS see Family Specifications.

Note

1. A destructive high current mode may occur if V_I and V_O are not constrained to the range $V_{SS} \leq V_I$ or $V_O \leq V_{DD}$.

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DC CHARACTERISTICS

 $V_{SS} = 0\text{ V}$

HEF	V_{DD} V	I_{OH} mA	SYMBOL	$T_{amb} (^{\circ}\text{C})$					
				-40		+ 25		+ 85	
				MIN.	MAX.	MIN.	TYP.	MIN.	MAX.
Output voltage HIGH	5	0	V_{OH}	4,10		4,10	4,40	4,10	V
	10	0		9,10		9,10	9,40	9,10	V
	15	0		14,10		14,10	14,40	14,10	V
Output voltage HIGH	5	5	V_{OH}				4,20		V
	10	5					9,20		V
	15	5					14,20		V
Output voltage HIGH	5	10	V_{OH}	3,60		3,60	4,05	3,30	V
	10	10		8,75		8,75	9,10	8,45	V
	15	10		13,75		13,75	14,10	13,45	V
Output voltage HIGH	5	15	V_{OH}				4,00		V
	10	15					9,00		V
	15	15					14,00		V
Output voltage HIGH	5	20	V_{OH}	2,80		2,80	3,80	2,50	V
	10	20		8,10		8,10	9,00	7,80	V
	15	20		13,10		13,10	14,00	12,80	V
Output voltage HIGH	5	25	V_{OH}				3,70		V
	10	25					8,90		V
	15	25					14,00		V

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HEC	V _{DD} V	I _{OH} mA	SYMBOL	T _{amb} (°C)					
				-55		+ 25		+ 125	
				MIN.	MAX.	MIN.	TYP.	MIN.	MAX.
Output voltage HIGH	5	0	V _{OH}	4,10		4,10	4,40	4,10	V
	10	0		9,10		9,10	9,90	9,10	V
	15	0		14,10		14,10	14,40	14,40	V
Output voltage HIGH	5	5	V _{OH}				4,30		V
	10	5					9,30		V
	15	5					14,30		V
Output voltage HIGH	5	10	V _{OH}	3,60		3,60	4,25	3,20	V
	10	10		8,75		8,75	9,25	8,35	V
	15	10		13,75		13,75	14,25	13,35	V
Output voltage HIGH	5	15	V _{OH}				4,20		V
	10	15					9,20		V
	15	15					14,20		V
Output voltage HIGH	5	20	V _{OH}	2,80		2,80	4,20	2,30	V
	10	20		8,10		8,10	9,20	7,60	V
	15	20		13,10		13,10	14,20	12,60	V
Output voltage HIGH	5	25	V _{OH}				4,15		V
	10	25					9,20		V
	15	25					14,20		V

AC CHARACTERISTICS

V_{SS} = 0 V; T_{amb} = 25 °C; input transition times ≤ 20 ns

	V _{DD} V	TYPICAL FORMULA FOR P (μW)	
Dynamic power dissipation per package (P)	5	$1\,000 f_i + \sum (f_o C_L) \times V_{DD}^2$	where f_i = input freq. (MHz) f_o = output freq. (MHz) C_L = load capacitance (pF) $\sum (f_o C_L)$ = sum of outputs V_{DD} = supply voltage (V)
	10	$4\,000 f_i + \sum (f_o C_L) \times V_{DD}^2$	
	15	$10\,000 f_i + \sum (f_o C_L) \times V_{DD}^2$	

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AC CHARACTERISTICS

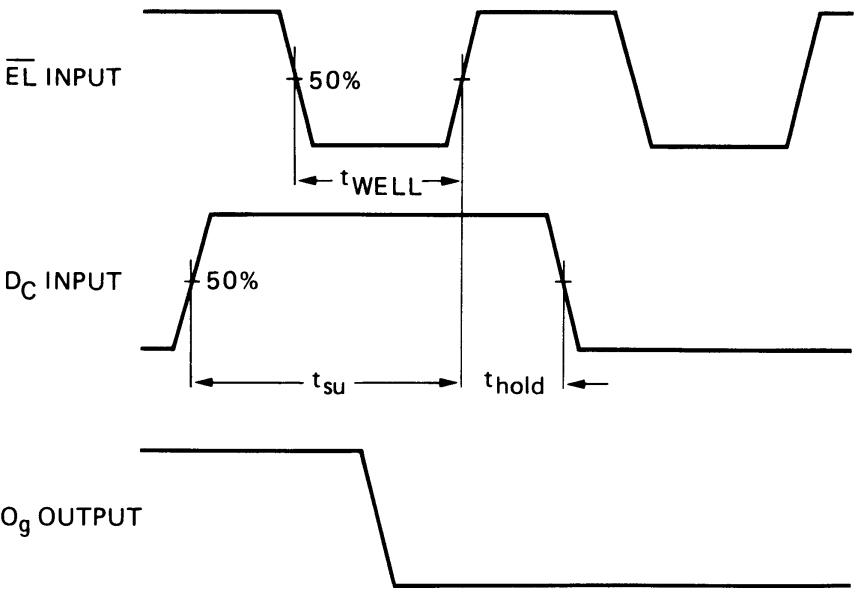
 $V_{SS} = 0\text{ V}$; $T_{amb} = 25\text{ }^{\circ}\text{C}$; $C_L = 50\text{ pF}$; input transition times $\leq 20\text{ ns}$

	V_{DD} V	SYMBOL	MIN.	TYP.	MAX.	TYPICAL EXTRAPOLATION FORMULA
Propagation delays						
$D_n \rightarrow O_n$	5			155	310 ns	128 ns + (0,55 ns/pF) C_L
HIGH to LOW	10	t_{PHL}		60	120 ns	49 ns + (0,23 ns/pF) C_L
	15			40	80 ns	32 ns + (0,16 ns/pF) C_L
LOW to HIGH	5	t_{PLH}		135	270 ns	108 ns + (0,55 ns/pF) C_L
	10			55	110 ns	44 ns + (0,23 ns/pF) C_L
	15			40	80 ns	32 ns + (0,16 ns/pF) C_L
$\overline{EL} \rightarrow O_n$	5			160	320 ns	133 ns + (0,55 ns/pF) C_L
HIGH to LOW	10	t_{PHL}		60	120 ns	49 ns + (0,23 ns/pF) C_L
	15			45	90 ns	37 ns + (0,16 ns/pF) C_L
LOW to HIGH	5	t_{PLH}		160	320 ns	133 ns + (0,55 ns/pF) C_L
	10			70	140 ns	59 ns + (0,23 ns/pF) C_L
	15			50	100 ns	42 ns + (0,16 ns/pF) C_L
$\overline{BI} \rightarrow O_n$	5			120	240 ns	93 ns + (0,55 ns/pF) C_L
HIGH to LOW	10	t_{PHL}		50	100 ns	39 ns + (0,23 ns/pF) C_L
	15			35	70 ns	27 ns + (0,16 ns/pF) C_L
$\overline{BI} \rightarrow O_n$	5			105	210 ns	78 ns + (0,55 ns/pF) C_L
LOW to HIGH	10	t_{PLH}		40	80 ns	29 ns + (0,23 ns/pF) C_L
	15			30	60 ns	22 ns + (0,16 ns/pF) C_L
$\overline{LT} \rightarrow O_n$	5			80	160 ns	52 ns + (0,55 ns/pF) C_L
HIGH to LOW	10	t_{PHL}		30	60 ns	19 ns + (0,23 ns/pF) C_L
	15			20	40 ns	12 ns + (0,16 ns/pF) C_L
LOW to HIGH	5	t_{PLH}		60	120 ns	33 ns + (0,55 ns/pF) C_L
	10			30	60 ns	19 ns + (0,23 ns/pF) C_L
	15			25	50 ns	17 ns + (0,16 ns/pF) C_L
Output transition times	5			60	120 ns	10 ns + (1,0 ns/pF) C_L
HIGH to LOW	10	t_{THL}		30	60 ns	9 ns + (0,42 ns/pF) C_L
	15			20	40 ns	6 ns + (0,28 ns/pF) C_L
LOW to HIGH	5	t_{TLH}		25	50 ns	20 ns + (1,0 ns/pF) C_L
	10			16	32 ns	13 ns + (0,06 ns/pF) C_L
	15			13	26 ns	10 ns + (0,06 ns/pF) C_L

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	V _{DD} V	SYMBOL	MIN.	TYP.	MAX.	TYPICAL EXTRAPOLATION FORMULA
Minimum $\overline{EL}$ pulse width; LOW	5	t_{WELL}	80	40	ns	see also waveforms Fig.8
	10		40	20	ns	
	15		35	17	ns	
Set-up time $D_n \rightarrow \overline{EL}$	5	t_{su}	50	25	ns	
	10		25	12	ns	
	15		20	9	ns	
Hold-time $D_n \rightarrow \overline{EL}$	5	t_{hold}	60	30	ns	
	10		30	15	ns	
	15		25	12	ns	



Conditions:
 $D_D = \text{LOW}$
 $D_A = D_B = \overline{B}_1 = \overline{L}\overline{T} = \text{HIGH}$

Fig.8 Waveforms showing minimum $\overline{EL}$ pulse width, set-up and hold time for D_C to $\overline{EL}$.

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APPLICATION INFORMATION

Some examples of applications for the HEF4511B are:

- Driving LED displays.
- Driving incandescent displays.
- Driving fluorescent displays.
- Driving LCD displays.
- Driving gas discharge displays.

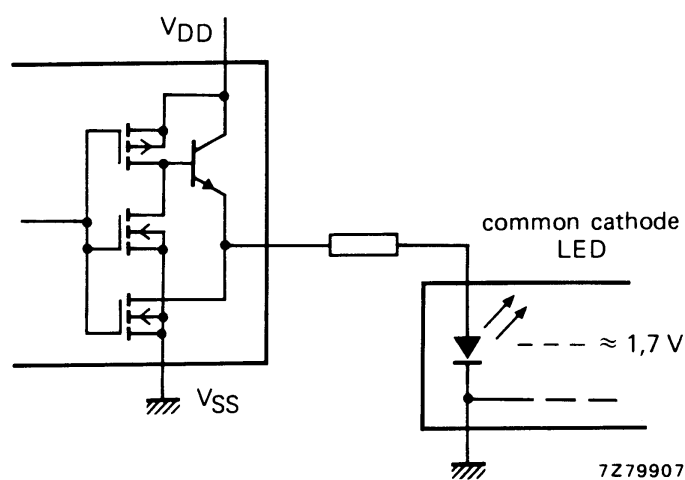


Fig.9 Connection to common cathode LED display readout.

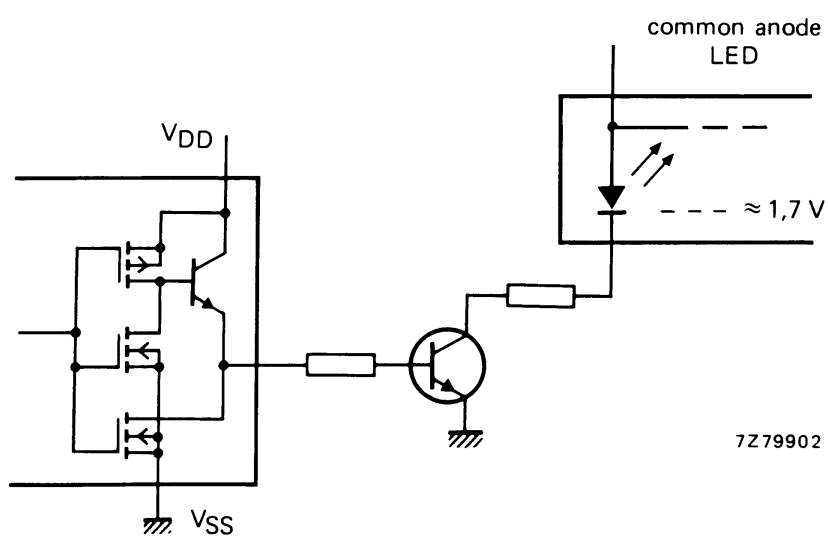


Fig.10 Connection to common anode LED display readout.

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